

Abdur Rahman Bin Shahid, Ph.D.

Assistant Professor and Director of SHIELD Lab
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RESEARCH INTEREST

My research focuses on enhancing the security, privacy, and trustworthiness of intelligent systems, with particular emphasis on developing sustainable secure AI, ensuring usable security and privacy in generative AI systems, exploring generative AI forensics techniques, and leveraging generative AI to improve the security and reliability of Cyber-Physical Systems.

EDUCATION

- **Ph.D.** in Computer Science Dec 2019
Florida International University (FIU), Miami, FL, USA
Dissertation: Trajectory Privacy-Preservation and Lightweight Blockchain Techniques for Internet of Things (IoT).
Advisor: Dr. Niki Pissinou
- **M.S.** in Computer Science Dec 2018
Florida International University (FIU), Miami, FL, USA
Advisor: Dr. Niki Pissinou
- **B.S.** in Computer Science and Engineering Dec 2011
Chittagong University of Engineering and Technology (CUET), Chittagong, Bangladesh.
Thesis: An image Watermarking Approach Based on Set Partitioning in Hierarchical Trees (SPIHT) Algorithm.
Advisor: Dr. Thomas Chowdhury

PROFESSIONAL EMPLOYMENT

- **Assistant Professor of Computer Science** Aug. 2023 - Present
Department of Computer Science, School of Computing, Southern Illinois University, Carbondale, IL, USA.
Director: Secure and Trustworthy Intelligent Systems (SHIELD) Lab
- **Assistant Professor of Computer and Information Systems** Aug. 2021 - Aug. 2023
Department of Computer and Information Systems, Robert Morris University, Moon Twp, PA, USA
- **Assistant Professor of Computer Science** Jan. 2020 - Aug. 2021
Department of Mathematics and Computer Science, Concord University, Athens, WV, USA
- **Graduate Assistant** Aug. 2014 - Dec. 2019
School of Computing and Information Sciences, Florida International University, Miami, FL, USA
- **Software Engineer** Jun. 2012 - Jul. 2014
Mobile Lab, Samsung R&D Institute Bangladesh (SRBD), Dhaka, Bangladesh
- **Software Engineer** Nov. 2012 - Jan. 2013
Samsung Electronics, Suwon, South Korea

AWARDS AND HONORS

- Summer Research Faculty Award, Undergraduate Summer Research Academy, Robert Morris University (2022).
- Dissertation Year Fellowship, Graduate School, Florida International University (2019).
- Best Graduate Student in Research, School of Computing and Information Sciences (SCIS), Florida International University (FIU) (2019).
- Best Poster Award (Runner-up), Undergraduate research category at Consortium for Computing Sciences in Colleges, (CCSC:Northeast). Supervised undergraduate researcher: Melinda Goda (2021).
- Graduate Assistantship, School of Computing and Information Sciences (SCIS), Florida International University (FIU) (2014-2018).

- Research Assistantship, School of Computing and Information Sciences (SCIS), Florida International University (FIU) (2015, 2019).

TEACHING EXPERIENCE

- **Assistant Professor of Computer Science** Aug 2023 - Present
Southern Illinois University
 - UHON351: AI Safety and Society (Spring 2026)
 - CS510: Wireless and Network Security (Spring 2026)
 - CS 409/CS 509: Ethical Hacking (Fall 2024, Fall 2025)
 - CS 413/CS 591 Digital Forensics (Fall 2023, Fall 2024)
 - CS 431: Cyber Physical Systems (Spring 2024, Spring 2025, Spring 2026)
 - CS 531: Security in Cyber Physical Systems (Spring 2025, Spring 2026)
- **Assistant Professor of Computer and Information Systems** Aug 2021 - Aug 2023
Robert Morris University
 - INFS 3152: Adv Java: Application Programming (Fall 2021, Spring 2022, Summer 2022, Fall 2022, Spring 2023)
 - INFS 4220: Data Mining Applications (Fall 2021, Fall 2022, Spring 2023)
 - INFS 4260/INFS 6730: Data Integration for Analytics / Data Warehousing (Fall 2021, Spring 2022, Fall 2022)
 - CYMS 6100: Foundations of Cybersecurity (Spring 2023)
 - INFS 3221: Advanced System Analysis and Design (Spring 2022, Summer 2022)
 - INFS 2150: Intro to Web development & E-Comm Tech (Spring 2022)
 - INFS 2020: Operating Systems (Spring 2023) (**Newly Developed**)
 - INFS 3500: Network Management and Administration (**Newly Developed**)
- **Assistant Professor of Computer Science** Jan 2020 - May 2021
Concord University
 - CS 221: Computer Networking (Spring 2020, Spring 2021)
 - CS 232: Introduction to Computer Security (Spring 2020, Spring 2021)
 - CS 261: Introduction to Intelligent Systems (Fall 2020)
 - CS 272: Human Centered Design (Fall 2020) (**Newly Developed**)
 - CS 357: Parallel and Distributed Computing (Spring 2021) (**Newly Developed**)
 - CS 374: Introduction to Simulation and Modeling (Spring 2020)
 - CS 421: Operating Systems (Spring 2020, Spring 2021)
 - CS 442: Analysis of Algorithms (Fall 2020)
 - CS 456: Capstone Project I (Fall 2020)
 - CS 457: Capstone Project II (Spring 2021)
- **Graduate Teaching Assistant** Aug 2014 - Dec. 2019
Florida International University
 - Worked as a TA for the following courses: Algorithms, Programming I, Programming II, Database Management, Introduction to Microcomputer Applications for Business, Principals of Programming Languages
 - Taught **Unreal Engine Game Development** and **Android Application Development** in NSF Research Experience for Teachers (RET) program at FIU.
 - Taught **Computational Privacy and Blockchain** in NSF Research Experience for Undergraduate (REU) program at FIU.

GRADUATE AND UNDERGRADUATE STUDENT SUPERVISION

- **Southern Illinois University**
Director, Secure and Trustworthy Intelligent Systems (SHIELD) Lab, School of Computing
 - **Doctoral Advisor**

- * Syed Mhamudul Hasan. *Topic: Sustainable Secure AI.*
- * Farhan Israk Soumik. *Topic: Usable Security and Privacy.*
- * Malithi Wanniarachchi Kankanamge (Co-Advising with Dr. Ning Yang). *Topic: Secure and Privacy-Preserving Generative AI.*
- **Master's Thesis & Project Advisor**
 - * Arun Aryal (Spring 2026 - Present) [**Thesis**]
 - * Bibek Koirala (Fall 2025 - Present) (Topic: Deep Learning Approach for Early Detection of Bacterial Contamination) [**Thesis**]
 - * bhargav vaddepally (Fall 2025) (*Topic: Deep Fake Detection Microservice*)
 - * Satwik Boyina (Spring 2023). *Topic: Wearble IoT Data Analytics.*
 - * Aman Singh Thakur (Spring 2023). *Topic: Wearable IoT testbed development.*
- **Undergraduate Research Advisor**
 - * Nadia Lafontant (Fall 2025 - Present) [**Reach Scholar**]
- **Robert Morris University**
 - **Undergraduate Research Advisor**
 - * Robert Schneeweis. *Topic: Testbed Development for Adversarial Attack for Smart Vehicles.*
 - * Ethan Bowman. *Topic: Testbed Development for Adversarial Attack for Smart Vehicles.*
 - * Marissa Beck. *Topic: Data Analytics and Cloud Computing Pedagogy.*
 - * Michael Depace. *Topic: Machine Learning Privacy.*
- **Concord University**
 - **Undergraduate Research Advisor**
 - * Melinda Goda. *Topic: Privacy-Preserving Machine Learning. (Runner-Up in Best Poster Award at CCSC: Northeast Conference 2021)*
 - * Diego Marti Perez. *Topic: Privacy-Preserving Machine Learning.*
 - * Kaylan Sheally. *Topic: Privacy-Preserving Machine Learning and Blockchain.*
 - * Aashish Poudel. *Topic: Privacy-Preserving Deep Learning.*
 - * Giovanni Quimbayo. *Topic: Privacy-Preserving Deep Learning.*
 - * Bevan Eckman. *Topic: Rank Prediction in Multiplayer games.*
- **NSF Research Experience for Undergraduates (NSF REU)**
School of Computing and Information Sciences,, Florida International University (FIU)
 - Worked as graduate mentor for undergraduate students.
 - * Edwin Aguilar. *Topic: Trajectory Privacy in Blockchain*
 - * Eric Perez. *Topic: Lightweight Blockchain*
 - * Sheila Alemany, FIU. *Topic: Trajectory Privacy in Blockchain.*
 - * Rain Kwan. *Topic: Lightweight Blockchain*
 - * Corey Staier. *Topic: Lightweight Blockchain*
 - * Liz Jeukeng. *Topic: Location Privacy Application Development*
 - * Aditya Shetty. *Topic: Computational Geometry in Cybersecurity*
 - * Keyan Zolfaghari. *Topic: Computational Geometry in Cybersecurity*
- **NSF Research Experience for Teachers (NSF RET)**
School of Computing and Information Sciences,, Florida International University (FIU)
 - Worked as graduate mentor for school teachers.
 - * Alejandra Penas. *Topic: Differential Privacy in Blockchain*
 - * Brook Still. *Topic: Differential Privacy for Location-Based Services*
 - * Teresita Lemus. *Topic: Geometry in Trajectory Privacy*
 - * Maite Varela-Conover. *Topic: Geometry in Trajectory Privacy*
 - * Jose Fernandez. *Topic: Computational Geometry in Cybersecurity*

OTHER WORK EXPERIENCE

- **Graduate Research Assistant**
Telecommunications & Information Technology Institute (IT2), SCIS, FIU

Aug. 2014 - Dec. 2019

- *Lightweight Spatiotemporal Data Privacy Preservation Technique Development*
 - * Researched on designing location privacy preserving techniques for a variety of location-based services: mobile Internet of Things (IoT) and Location-Based Social Network (LBSN).
 - * Invented a technique for applying geometric structure for location privacy preservation from the theory of Voronoi diagram.
 - * Designed a delay aware continuous location privacy preservation technique by proposing a balance between time-based quality of service and degree of privacy.
 - * Devised a location inference attack model to infer user's location information at a higher level through the analysis of spatiotemporal correlation between different events in LBSN (e.g. check-ins and photos).
 - *Lightweight Blockchain for Mobile Internet of Things (IoT)*
 - * Invented *Sensor-Chain*, an innovative spatiotemporal correlation-driven approach to make blockchain lightweight for resource-constrained sensing devices.
 - * Assisted my PhD advisor in writing successful grants, including *Sensor-Chain: A Demo of a Lightweight Blockchain Based Internet of Military Things Security Scheme*, a grant funded by the U.S. Air Force Research Laboratory.
 - Collaborated with other researchers on Unmanned Ariel Vehicle (UAV), trust-based crowdsensing data cleaning, and 3D computer graphics projects.
 - Collaboratively developed the first version of GEOM SE, a repository for 3D geometric objects with search and visualization functionalities.
- **Graduate Mentor** May. 2015 - Jul. 2019
 NSF Research Experience for Undergraduates (REU) and NSF Research Experience for Teachers (RET), SCIS, FIU
 - Mentored undergraduate students and K-12 teachers at REU and RET sites in research on blockchain, IoT, privacy, and computational geometry in cybersecurity.
 - Ranked among the top 5% of graduate mentors for publishing research co-authored with REU and RET participants in top conferences.
 - **Software Engineer (Jun. 2012 - Jul. 2014)**
 Mobile Lab, Samsung R&D Institute Bangladesh (SRBD), Dhaka, Bangladesh
Responsibilities and Achievements
 - Designed, developed, and maintained Homescreen and Accuweather applications in Samsung's Android phones and tablets for the Middle East and Africa region's telecom vendors.
 - Proposed augmented reality-based unlocking system for head mounted display devices.
 - **Software Engineer (Nov. 2012 - Jan. 2013)**
 Samsung Electronics, Suwon, South Korea
Responsibilities and Achievements
 - Refactored the homescreen application of Samsung S4 device and reduced its rendering delay to 3 seconds.
 - Researched on visualization of homescreen background image without spanning multiple panels.

DISSERTATION AND THESIS

- **Abdur R. Shahid**, "Trajectory Privacy Preservation and Lightweight Blockchain Techniques for Mobility-Centric IoT" (2019). FIU Electronic Theses and Dissertations.
- **Abdur R. Shahid**, "An image Watermarking Approach Based on Set Partitioning in Hierarchical Trees (SPIHT) Algorithm" (2011), Chittagong University of Engineering and Technology (CUET).

BOOK PUBLICATIONS

- **Abdur R. Shahid**, Niki Pissinou, and Sajedul Talukder. "Protecting Location Privacy in Blockchain-based Mobile Internet of Things". In: Daimi, K., Dionysiou, I., El Madhoun, N. (eds) Principles and Practice of Blockchains, Springer, USA 2022.

JOURNAL PUBLICATIONS

★ Supervised Ph.D. Student

- Syed Mhamudul Hasan* and **Abdur R. Shahid**, "Generative AI as Survey Participants: A Human-Centered Evaluation of LLM Behavioral Response", IEEE Intelligent Systems (In Review)
- Malithi Mithsara Wanniarachchi Kankanamge*, Ning Yang, Hussein Zangoti, Ahmed Imteaj, and **Abdur R. Shahid**, "Adaptive and Robust Data Poisoning Detection and Sanitization in Wearable IoT Systems using Large and Small Language Models", IEEE Transactions on Human-Machine Systems (In Review)
- Malithi Mithsara Wanniarachchi Kankanamge*, Syed Mhamudul Hasan, Hussein Zangoti, Ahmed Imteaj, Ning Yang, and **Abdur R. Shahid**, "A Comprehensive Survey of Large Language Models for Human Activity Recognition: Opportunities, Challenges, and Future Directions", IEEE Transactions on Emerging Topics in Computational Intelligence (In Review)
- Awal Ahmed Fime, Tasfia Zaman Samiha, Md Zarif Hossain, Saika Zaman, Ashfak Md Shibli, **Abdur R. Shahid**, Zhen Ni, and Ahmed Imteaj, "Secure yet fragile: adversarial vulnerabilities of federated vision-language models in medical AI". Scientific Reports 16, 17701 (2026). <https://doi.org/10.1038/s41598-026-48102-4>
- Syed Mhamudul Hasan*, Ahmed Imteaj, and **Abdur R. Shahid**, "Quantifying Robustness and Sustainability Trade-off in Federated Adversarial Learning for Cyber-Physical Systems", IEEE Transactions on Sustainable Computing (2026)
- Syed Mhamudul Hasan*, Taminul Islam, Munshi Saifuzzaman, Khaled R Ahmed, Chun-Hsi Huang, and **Abdur R. Shahid**, "Carbon Emission Quantification of Machine Learning: A Review", IEEE Transactions on Sustainable Computing (2025)
- Syed Mhamudul Hasan*, **Abdur R. Shahid**, and Ahmed Imteaj, "Evaluating Sustainability and Social Costs of Adversarial Training in Machine Learning", In IEEE Consumer Electronics Magazine (2024).
- **Abdur R. Shahid** and Sushma Mishra, "A Framework for A Masters in Applied Artificial Intelligence Program in Computer and Information Systems Discipline", Journal of Information Systems Education (2024).
- **Abdur R. Shahid** and Ahmed Imteaj, "Securing User Privacy in Cloud-Based Whiteboard Services Against Health Attribute Inference Attacks", in IEEE Transactions on Artificial Intelligence (2024).
- **Abdur R. Shahid** and Ahmed Imteaj, "Sticks and Stones May Break My Bones, But Words Will Never Hurt Me! — Navigating the Cybersecurity Risks of Generative AI", AI & Soc (2024).
- Ahmed Imteaj, **Abdur R. Shahid**, and Saika Zaman, "Leveraging Blockchain Interoperability for Interdependent Networks", In IEEE Consumer Electronics Magazine (2023).
- **Abdur R. Shahid**, Niki Pissinou, S. S. Iyengar, and Kia Makki. "Delay-aware privacy-preserving location-based services under spatiotemporal constraints." International Journal of Communication Systems (2020): e4656.
- Zeng, Wei, Muhammad Razib, and **Abdur R. Shahid**. "Diffeomorphism Spline." Axioms 4, no. 2 (2015): 156-176.

CONFERENCE PUBLICATIONS

★ Supervised Ph.D. Student.

- Wanniarachchi Kankanamge Malithi Mithsara*, Ning Yang, Chun-Hsi Huang, Hussein Zangoti, **Abdur R. Shahid**, "The Little Models That Think: A Review of Reasoning in Small Language Models", The 20th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2027) (In Review).
- Mohd. Farhan Israk Soumik*, Awal Ahmed Fime, Wanniarachchi Kankanamge Malithi Mithsara, Arun Babu Aryal, Hoda Joudi Haghighi, Bibek Koirala, Ahmed Imteaj, **Abdur R. Shahid**, "Edits in the Wild: A Multilingual Benchmark for Localizing On-Device Generative Image Edits on Consumer Platforms", The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026) (In Review).

- Syed Mhamudul Hasan, Mohd. Farhan Israk Soumik, Hussein Zangoti*, Ahmed Imteaj and Abdur R. Shahid, “QPrint: Side-Channel Fingerprinting of Quantum AI Workloads across Heterogeneous Devices,” The IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2026) (In Review)
- Mohd Farhan Israk Soumik, Hussein Zangoti, Arun Babu Aryal, Ahmed Imteaj and Abdur R. Shahid, “When Quantum Entanglement Meets Transformer Attention: A Controlled Study of Quantum Attention Heads” The IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2026) (In Review)
- Mohd. Farhan Israk Soumik*, Syed Mhamudul Hasan, WKM. Mithsara, Ahmed Imteaj, **Abdur R. Shahid**, “No Innocence in Styling: Discovery of Privacy Protection Capabilities and Security Risks in Consumer Generative AI Writing Assistants”, In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track), pages 1866–1885, San Diego, California, USA. Association for Computational Linguistics.
- Syed Mhamudul Hasan*, Hussein Zangoti, Iraklis Anagnostopoulos, **Abdur R. Shahid**, “Sponge Attacks on Sensing AI: Energy-Latency Vulnerabilities and Defense via Model Pruning”, In the 2025 IEEE Global Communications Conference (GLOBECOM), 2025.
- Mohd Farhan Israk Soumik*, Ebrahim Maghsoudlou Nima, Fime, A. Imteaj, **Abdur R. Shahid**, “Privacy Preserving Multimodal Stress Detection from Wearables with Attention Fusion and Federated Learning”, In Proceedings of the 24th International Conference on Machine Learning Applications (ICMLA 2025)
- WKM Mithsara*, **Abdur R. Shahid**, Ning Yang, SM Jamil Uddin, and Rahul Biswas, “Development of a Fall Detection and Safety Communication System Using Small Language Models”, In The 42nd International Symposium on Automation and Robotics in Construction (ISARC 2025).
- SM Jamil Uddin, **Abdur R. Shahid**, Mohd Farhan Israk Soumik, and Ziyu Jin, “Development of a Hand-Arm Vibration Syndrome (HAVS) Detection Tool for Construction Workers”, In The 42nd International Symposium on Automation and Robotics in Construction (ISARC 2025).
- Nishi Chaudhary, SM Jamil Uddin, Mahzabin Tahmanna, Alex Albert, and **Abdur R. Shahid**, “How Reliable Are Large Language Models? Zero-Shot Detection of Construction Hazards”, In The 42nd International Symposium on Automation and Robotics in Construction (ISARC 2025).
- WKM Mithsara*, **Abdur R. Shahid**, and Ning Yang, “Zero-Shot Detection and Sanitization of Data Poisoning Attacks in Wearable AI using Large Language Models”, In The 23rd IEEE International Conference on Machine Learning and Applications (ICMLA 2024).
- WKM Mithsara*, **Abdur R. Shahid**, and Ning Yang, “Intelligent Fall Detection and Emergency Response for Smart Homes Using Language Models”, In The 23rd IEEE International Conference on Machine Learning and Applications (ICMLA 2024).
- Awal Ahmed Fime, Md Zarif Hossain, Saika Zaman, **Abdur R. Shahid**, and Ahmed Imteaj. “Benchmarking Large-Language Models for Resource-Efficient Medical AI for Edge Deployment.” In Proceedings of the AAAI Symposium Series, vol. 5, no. 1, pp. 137-142. 2025.
- Md Zarif Hossain, **Abdur R. Shahid**, and Ahmed Imteaj, “Breaking and Securing Vision-Language Models: An AdversarialRobustness Study”, In Proceedings of the 24th International Conference on Machine Learning Applications (ICMLA 2025)
- Ahmed Imteaj, Md Zarif Hossain, Saika Zaman, **Abdur R. Shahid**, “TriplePlay: Enhancing Federated Learning with CLIP for Non-IID Data and Resource Efficiency”, In The 23rd IEEE International Conference on Machine Learning and Applications (ICMLA 2024).
- Mohammad Imran Hossain, **Abdur R. Shahid**, Ahmed Imteaj, “A Diffusion of Innovation-Driven Approach for Strategic Integration of ChatGPT in Education”, ISCAP Conference (2024).
- **Abdur R. Shahid**, Syed Mhamudul Hasan*, Malithi Wanniarachchi Kankanamge*, Md Zarif Hossain and Ahmed Imteaj, “WatchOverGPT: A Framework for Real-Time Crime Detection and Response Using Wearable Camera and Large Language Model (LLM)”, In The 48th IEEE International Conference on Computers, Software, and Applications (COMPSAC 2024).

- Ahmed Imteaj, Tanveer Rahman, Saika Zaman, Md Zarif Hossain and **Abdur R. Shahid**, “Enhancing Road Safety through Cost-Effective, Real-Time Monitoring of Driver Awareness with Resource-Constrained IoT Devices”, In The 48th IEEE International Conference on Computers, Software, and Applications (COMPSAC 2024).
- Syed Mhamudul Hasan*, Alaa M Alotaibi, Sajedul K Talukder and **Abdur R. Shahid**, “Distributed Threat Intelligence at the Edge Devices: A Large Language Model (LLM)-Driven Approach”, In the 48th IEEE International Conference on Computers, Software, and Applications (COMPSAC 2024).
- WKM Malithi*, **Abdur R. Shahid**, and Ning Yeng, “A Large Language Models (LLM)-Integrated Healthcare Cyber-Physical Systems Architecture”, In the 48th IEEE International Conference on Computers, Software, and Applications (COMPSAC 2024).
- Oleksandr Jockusch, Md Zarif Hossain, Ahmed Imteaj, **Abdur R. Shahid**, “Generative AI-Based Land Cover Classification via Federated Learning CNNs: Sustainable Insights from UAV Imagery”, In The 2024 IEEE Conference on Technologies for Sustainability (SusTech 2024).
- Syed Mhamudul Hasan*, **Abdur R. Shahid**, Ahmed Imteaj, “The Environmental Price of Intelligence: Evaluating the Social Cost of Carbon in Machine Learning”, In The 2024 IEEE Conference on Technologies for Sustainability (SusTech 2024)
- Md Zarif Hossain, Imteaj Ahmed, Saika Zaman, **Abdur R. Shahid**, Sajedul Talukder, MH Amini, “FLID: Intrusion Attack and Defense Mechanism for Federated Learning-Empowered Connected Autonomous Vehicles”, In The 2023 6th IEEE Conference on Dependable and Secure Computing (IEEE DSC 2023)
- **Abdur R. Shahid**, and Ahmed Imteaj, “Assessing Wearable Human Activity Recognition Systems Against Data Poisoning Attacks in Differentially-Private Federated Learning.” In The 2023 IEEE International Conference on Smart Computing (SMARTCOMP). IEEE, 2023.
- **Abdur R. Shahid**, Ahmed Imteaj, Peter Y. Wu, Diane A. Igoche, Tauhidul Alam, “Label Flipping Data Poisoning Attack Against Wearable Human Activity Recognition System”. In 2022 IEEE Symposium Series on Computational Intelligence (IEEE SSCI), 2022.
- Hussein Zangoti, Alex P. Makki, Niki Pissinou, **Abdur R. Shahid**, Omar J. Guerra, Joel Rodriguez, “A Multidimensional Blockchain Framework For Mobile Internet of Things”. In 2022 International Conferences on Trust, Security and Privacy in Computing and Communications (IEEE TrustCom), IEEE, 2022.
- **Abdur R. Shahid**, “Facilitating Internet of Things (IoT) Experience in Computer and Information Systems Education”, In proceedings of the 38th ACM CCSC Eastern Conference (ACM CCSC), Oct 2022.
- **Abdur R. Shahid**, and Sajedul Talukder, “Evaluating Machine Learning Models for Handwriting Recognition-based Systems under Local Differential Privacy”, The Conference on Innovations and Applications in Intelligent Systems (ASYU), 2021.
- **Abdur R. Shahid**, and Sajedul Talukder, “A Study of Differentially Private Machine Learning in Healthcare”, The Conference on Innovations and Applications in Intelligent Systems (ASYU), 2021.
- Sajedul Talukder and **Abdur R. Shahid**, “Applying Local Differential Privacy in Handwriting Recognition-based Systems”, In proceedings of the 37th ACM CCSC Eastern Conference (ACM CCSC), Oct 2021.
- Sajedul Talukder and **Abdur R. Shahid**, “Enhancing Alternative Proof of Work for Cryptocurrencies Using Machine Learning”, In proceedings of the 37th ACM CCSC Eastern Conference (ACM CCSC), Oct 2021.
- **Abdur R. Shahid** and Sajedul Talukder, “Privacy-Preserving Activity Recognition from Sensor Data”, In proceedings of the 37th ACM CCSC Eastern Conference (ACM CCSC), Oct 2021.
- **Abdur R. Shahid** and Sajedul Talukder, “Evaluation of Privacy-Preserving Logistic Regression and Naive Bayes Classifiers in Healthcare”, In proceedings of the 37th ACM CCSC Eastern Conference (ACM CCSC), Oct 2021.
- T. Alam, J. Taylor, J. Taylor, S. Badsha, **Abdur R. Shahid**, A. S. M. Kayes, “Leveraging Blockchain for Spoof-Resilient Robot Networks, 13th International Conference on Intelligent Robotics and Applications” (ICIRA 2020), LNCS, Springer, 2020.

- Arpan Bhattacharjee, Shahriar Badsha, **Abdur R. Shahid**, Hanif Livani, Shamik Sengupta, “Block-Phasor: A Decentralized Blockchain Framework to Enhance Security of Synchrophasor”, 2020 IEEE Kansas Power and Energy Conference (KPEC), 2020.
- **Abdur R. Shahid**, Niki Pissinou, laurent Njilla, Edwin Aguilar*, and Eric Perez*. “Demo: Towards the Development of a Differentially Private Lightweight and Scalable Blockchain for IoT.” In 16th IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS), IEEE, 2019.
- **Abdur R. Shahid**, Niki Pissinou, Sheila Alemany, Ahmed Imteaj, Kia Makki, and Laurent Njilla. “Quantifying Location Privacy in Permissioned Blockchain-based Internet of Things (IoT)”. In EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services (MobiQuitous), 2019.
- **Abdur R. Shahid**, Niki Pissinou, Corey Staier*, and Rain Kwan*. “Sensor-Chain: A Lightweight Scalable Blockchain Framework for Internet of Things”. In International Conference on Internet of Things (iThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCom) and IEEE Smart Data (SmartData) (pp. 1154-1161). IEEE, 2019.
- Tang, Yujian, Samia, Tasnim, Niki Pissinou, S. S. Iyengar, and **Abdur R. Shahid**. “Reputation-Aware Data Fusion and Malicious Participant Detection in Mobile Crowdsensing.” In 2018 IEEE International Conference on Big Data (Big Data), pp. 4820-4828. IEEE, 2018.
- **Abdur R. Shahid**, Niki Pissinou, S. S. Iyengar, and Kia Makki. “Check-ins and Photos: Spatiotemporal Correlation-Based Location Inference Attack and Defense in Location-Based Social Networks.” In 2018 17th IEEE International Conference On Trust, Security And Privacy In Computing And Communications/12th IEEE International Conference On Big Data Science And Engineering (TrustCom/BigDataSE), pp. 1852-1857. IEEE, 2018.
- **Abdur R. Shahid**, Niki Pissinou, S. S. Iyengar, Jerry Miller, Ziqian Ding, and Teresita Lemus. “KLAP for Real-World Protection of Location Privacy.” In 2018 IEEE World Congress on Services (SERVICES), pp. 17-18. IEEE, 2018.
- **Abdur R. Shahid**, Liz Jeukeng*, Wei Zeng, Niki Pissinou, S. S. Iyengar, Sartaj Sahni, and Maite Varela-Conover. “PPVC: Privacy Preserving Voronoi Cell for location-based services.” In 2017 International Conference on Computing, Networking and Communications (ICNC), pp. 351-355. IEEE, 2017.

WORKSHOP AND POSTER PUBLICATIONS

- Malithi Mithsara Wanniarachchi Kankanamge*, Ning Yang, Ahmed Imteaj, **Abdur R. Shahid**, “Breaking Sensor-Text Alignment: Cross-Modal Attack on Contrastive Multimodal Human Activity Recognition”. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPRW 2026).
- Farhan Israk Soumik*, WKM Mithsara*, **Abdur R. Shahid**, Ahmed Imteaj, “Exploring Audio Editing Features as User-Centric Privacy Defenses Against Emotion Inference Attacks”, The Sixth AAAI-25 Workshop on Privacy-Preserving Artificial Intelligence (PPAI-AAAI), 2025.
- WKM Mithsara*, **Abdur R. Shahid**, and Ning Yang, “Leveraging Large Language Models for Zero-Shot Detection and Mitigation of Data Poisoning in Wearable AI Systems”, NeurIPS workshop on GenAI for Health: Potential, Trust and Policy Compliance (GenAI-NeurIPS), 2024.
- Syed Mhamudul Hasan*, **Abdur R. Shahid**, Ahmed Imteaj, and Hussein Zangoti, “Energy-Latency Sponge Attack on Wearable AI”, In The IEEE-EMBS International Conference on Body Sensor Networks (IEEE BSN) 2024.
- WKM Mithsara*, **Abdur R. Shahid**, and Ning Yang, “Detection of Poisoning Attacks in Human Activity Recognition using Large Language Models”, In The IEEE-EMBS International Conference on Body Sensor Networks (IEEE BSN) 2024.
- **Abdur R. Shahid**, Syed Mhamudul Hasan*, Ahmed Imteaj, Shahriar Badsha, “Poster: Context-Aware Spatiotemporal Poisoning Attacks on Wearable-Based Activity Recognition”, In The IEEE International Conference on Computer Communications (IEEE INFOCOM), 2024.

- Syed Mhamudul Hasan*, **Abdur R. Shahid**, and Ahmed Imteaj “Towards Sustainable SecureML: Quantifying Carbon Footprint of Adversarial Machine Learning” In The GreenNet Workshop at IEEE International Conference on Communications (IEEE ICC), 2024.
- Melinda Goda*, and **Abdur R. Shahid**, “Differentially Private Machine Learning for Breast Cancer Classification.” 25th Annual Conference of the Consortium for Computing Sciences in Colleges, Northeastern Region (CCSCNE), 2021. (**Runner-up in undergraduate best poster competition**).
- Diego Perez*, **Abdur R. Shahid** “Privacy-Preserving Human Activity Recognition Using Differentially Private Machine Learning” Consortium for Computing Sciences in Colleges, CCSC:Central Plains, 2021.
- Kaylan Sheally*, **Abdur R. Shahid** “Development of a Simulator for Blockchain Framework to Enhance Security and Privacy in Autonomous Robots in Adverse Environment” Consortium for Computing Sciences in Colleges, CCSC:Central Plains, 2021.

PREPRINTS

- Syed Mhamudul Hasan*, Anas AlSobeh, Hussein Zangoti, **Abdur R. Shahid**. “VetClaw: An Edge-Cloud Multimodal Agentic System for Veterinary Disease Screening”. arXiv preprint arXiv:2607.26042 (2026)
- Syed Mhamudul Hasan*, **Abdur R. Shahid**. “Modeling Emotional Dynamics in Agent-to-Agent Interactions on Moltbook”. arXiv preprint arXiv:2605.20442 (2026)
- Syed Mhamudul Hasan*, **Abdur R. Shahid**. “Emotion-Aware Clickbait Attack in Social Media”. arXiv preprint arXiv:2604.27369 (2026)
- Mohd Farhan Israk Soumik*, Syed Mhamudul Hasan*, and **Abdur R. Shahid**. “Evaluating Apple Intelligence’s Writing Tools for Privacy Against Large Language Model-Based Inference Attacks: Insights from Early Datasets.” arXiv preprint arXiv:2506.03870 (2025).
- Kankanamege, Malithi Wanniarachchi*, Nick McKenna, Santiago Carmona, Syed Mhamudul Hasan*, Ahmed Imteaj, and Abdur R. Shahid. ”Digital Forensic Investigation of the ChatGPT Windows Application.” arXiv preprint arXiv:2505.23938 (2025).
- Zeng, Wei, **Abdur R. Shahid**, Keyan Zolfaghari, Aditya Shetty, Niki Pissinou, and Sitharama S. Iyengar. “ n -VDD: Location Privacy Protection Based on Voronoi-Delaunay Duality.” arXiv preprint arXiv:1906.09158 (2019).

PRESENTATIONS

- “Zero-Shot Detection and Sanitization of Data Poisoning Attacks in Wearable AI using Large Language Models”, Large Language Models (LLM) Nexus: Bridging Technical Innovation and Ethical Horizons (SIU), 2025.
- “Lightweight Blockchain for Resource-Constrained IoT”. International E-Conference on Computer Networks and Soft Computing, ICCNSC 2020.
- “AI for Security and Security for AI”. Hopper Turing Society. Concord University. 2020
- “Computer Science Minor Program”. Major Minor Fair, Concord University, 2020.
- “Trajectory Privacy Preservation and Lightweight Blockchain Techniques for Mobility-Centric IoT”. Florida International University, 2019.
- “Check-ins and Photos: Spatiotemporal Correlation-Based Location Inference Attack and Defense in Location-Based Social Networks.”. Conference presentation at IEEE Trustcom 2018.
- “KLAP for Real-World Protection of Location Privacy”. Conference presentation at IEEE SERVICES, 2018.

PROFESSIONAL SERVICES

Proposal Reviewer

- Directorate for STEM Education (EDU), National Science Foundation (NSF) (2025)

Editorial Board Member

- Sensor. Guest editor of special issue on “IoT and Artificial Intelligence Approaches to Defeat COVID-19 Outbreak”.
- Topic Editor for the research topic “Security Breaches and Access Control” (RT ID:11427), Frontiers.

Program Committee Member

- TPC member of IEEE Consumer Communications & Networking Conference (IEEE CCNC) (2023, 2024, 2025).
- TPC member of IEEE Frontiers in Education (FIE) (2024).
- TPC member of IEEE Symposium Series on Computational Intelligence (IEEE SSCI) (2022-2023).
- Co-chair of the steering committee, International Workshop on Security, Privacy, and Trust for Emergency Events (EmergencyComm) in conjunction with the 16th EAI International Conference on Security and Privacy in Communication Networks (SecureComm 2020) conference, 2020.
- Session chair, “Information theory & IOT & Data Science” session in International IOT, Electronics and Mechatronics Conference (IEMTRONICS 2020).

Journal Reviewer

- Elsevier Journal of Engineering Research
- IEEE Internet of Things
- IEEE Transactions on Industrial Informatics
- IEEE transactions on consumer electronics
- ACM Transactions on Internet of Things (2023-)
- Current Research in Behavioral Sciences
- Frontiers in Computer Science.
- Engineering Applications of Artificial Intelligence - Elsevier.
- IEEE Journal of Biomedical and Health Informatics.
- IEEE Transactions on Dependable and Secure Computing
- International Journal of Communication Systems, Wiley.
- Sensors, Electronics, and AI - MDPI.
- Computer Communications - Elsevier.

Conference Reviewer

- IEEE Frontiers in Education (FIE), 2024.
- ISCAP Conference, 2024.
- IEEE Consumer Communications & Networking Conference (IEEE CCNC), 2023.
- IEEE Symposium Series on Computational Intelligence (IEEE SSCI), 2022.
- IEEE International Conference on Distributed Computing Systems (ICDCS 2019).
- Consortium for Computing Sciences in Colleges — Northeastern Region (CCSCNE 2022).

Academic Service

- Southern Illinois University Carbondale
 - Tenure Track Faculty Search Committee - AI (2025, 2026)
 - None Tenure Track Faculty Search Committee - AI (2026)
 - Course syllabus development for new BS in AI+ program (2025):
 - * AIP 290: Ethics, Safety and Security of AI
 - * AIP 370: Introduction to Prompt Engineering
- Robert Morris University
 - CIS Assessment Committee (2022 - 2023).

- B.S. in Computer Science Program Development Committee. (2022 - 2023).
- CIS Faculty Search Committee (2021-2022).
- Data Science core faculty (2022-2023).
- CIS faculty Representative at RMU Scholars Day, Discovery Day, Graduate Admission event, and Prospective student meeting (2021-2023).

- **Concord University**

- CS program Assessment Committee (2020-2021).
- CS minor coordinator (2020-2021).
- Veterans Liaison program (2020-2021).